

Numerical Methods for PDEs - Exam Cheat Sheet

WEEK 1: PDE FUNDAMENTALS

Method of Characteristics

Advection Equation: $\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = 0$

Solution: $u(x, t) = f(x - at)$

- Waves travel at speed a - Characteristic lines: $\frac{dx}{dt} = a$

Fourier Analysis Recipe

1. Assume: $u(x, t) = e^{i(kx + \omega t)}$
2. Compute: $\frac{\partial u}{\partial t} = i\omega \cdot u$, $\frac{\partial u}{\partial x} = ik \cdot u$
3. Substitute into PDE
4. Solve for **dispersion relation:** $\omega(k)$

Key Dispersion Relations:

- **Advection:** $\omega = -ak$ (real $\rightarrow$ propagation)
- **Diffusion:** $\omega = i\nu k^2$ (imaginary $\rightarrow$ decay)

PDE Classification (2nd Order)

Form: $Au_{xx} + Bu_{xy} + Cu_{yy} + \dots = 0$

Discriminant: $\Delta = B^2 - 4AC$

- $\Delta > 0$: **Hyperbolic** (wave equation)
- $\Delta = 0$: **Parabolic** (heat equation)
- $\Delta < 0$: **Elliptic** (Laplace equation)

Conservation Laws

Form: $\frac{\partial \rho}{\partial t} + \frac{\partial F}{\partial x} = 0$

Conservation: $\int \rho dx = \text{constant}$ if $F \rightarrow 0$ as $|x| \rightarrow \infty$

Stability (Fourier Space)

ODE: $\frac{d\hat{u}}{dt} = A(ik)\hat{u}$

Solution: $\hat{u}(k, t) = e^{A(ik)t}\hat{u}(k, 0)$

Stability Criterion:

- $\text{Re}(A(ik)) < 0 \rightarrow$ **Stable** (decay)
- $\text{Re}(A(ik)) = 0 \rightarrow$ **Neutral**
- $\text{Re}(A(ik)) > 0 \rightarrow$ **Unstable** (growth)

Magnitude: $|e^{A(ik)t}| = e^{\text{Re}(A(ik)) \cdot t}$

WEEK 2: FINITE DIFFERENCES

Grid Notation

- $x_j = j \cdot \Delta x$ for $j = 0, 1, \dots, N$ - $t_n = n \cdot \Delta t$ for $n = 0, 1, \dots, M$
- $u_j^n = u(x_j, t_n)$

First Derivative Approximations

Forward Difference (1st order):

$$\left. \frac{\partial u}{\partial x} \right|_j \approx \frac{u_{j+1} - u_j}{\Delta x}, \quad \text{Error} = O(\Delta x)$$

Backward Difference (1st order):

$$\left. \frac{\partial u}{\partial x} \right|_j \approx \frac{u_j - u_{j-1}}{\Delta x}, \quad \text{Error} = O(\Delta x)$$

Central Difference (2nd order):

$$\left. \frac{\partial u}{\partial x} \right|_j \approx \frac{u_{j+1} - u_{j-1}}{2\Delta x}, \quad \text{Error} = O(\Delta x^2)$$

Second Derivative

Central (Compact) (2nd order):

$$\left. \frac{\partial^2 u}{\partial x^2} \right|_j \approx \frac{u_{j+1} - 2u_j + u_{j-1}}{\Delta x^2}$$

5-point stencil (4th order):

$$\left. \frac{\partial^2 u}{\partial x^2} \right|_j \approx \frac{-u_{j+2} + 16u_{j+1} - 30u_j + 16u_{j-1} - u_{j-2}}{12\Delta x^2}$$

Taylor Series

$$u(x + \Delta x) = u(x) + \Delta x u'(x) + \frac{\Delta x^2}{2!} u''(x) + \frac{\Delta x^3}{3!} u'''(x) + \dots$$

Error Norms

$$\|e\|_\infty = \max_j |e_j| \quad (\text{infinity norm})$$

$$\|e\|_1 = \sum_j |e_j| \quad (1\text{-norm})$$

$$\|e\|_2 = \sqrt{\sum_j e_j^2} \quad (2\text{-norm, RMS})$$

Consistency

FD approximation is **consistent** if error $\rightarrow 0$ as $\Delta x \rightarrow 0$

Order of Accuracy: $E(\Delta x) \approx C \cdot \Delta x^p$

WEEK 3: TIME INTEGRATION

Euler Explicit (Forward Euler)

$$u^{n+1} = u^n + \Delta t \cdot L(u^n)$$

- **Error:** $O(\Delta t)$
- **Stability:** CONDITIONAL

Euler Implicit (Backward Euler)

$$u^{n+1} - \Delta t \cdot L(u^{n+1}) = u^n$$

- **Error:** $O(\Delta t)$
- **Stability:** UNCONDITIONAL (A-stable)

Trapezoidal Method

$$u^{n+1} = u^n + \frac{\Delta t}{2} [L(u^n) + L(u^{n+1})]$$

- **Error:** $O(\Delta t^2)$
- **Stability:** UNCONDITIONAL (A-stable)

Amplification Factor

$$G(z) = \frac{u^{n+1}}{u^n}, \quad z = \lambda \Delta t$$

Stability: $|G(z)| \leq 1$

Common Factors:

$$\text{Euler Explicit: } G = 1 + z$$

$$\text{Euler Implicit: } G = \frac{1}{1 - z}$$

$$\text{Trapezoidal: } G = \frac{1 + z/2}{1 - z/2}$$

Stability Regions

Euler Explicit:

$$|1 + z| \leq 1 \implies -2 \leq \text{Re}(z) \leq 0$$

A-Stability: Entire left half-plane ($\text{Re}(z) < 0$)

- Euler Implicit: A-stable ✓
- Trapezoidal: A-stable ✓
- Euler Explicit: NOT A-stable ✗

Heat Equation Stability

Euler Explicit Scheme:

$$u_j^{n+1} = u_j^n + r(u_{j+1}^n - 2u_j^n + u_{j-1}^n)$$

where $r = \frac{\nu \Delta t}{\Delta x^2}$

Stability: $r \leq \frac{1}{2} \implies \Delta t \leq \frac{\Delta x^2}{2\nu}$

WEEK 4: RUNGE-KUTTA

RK2 (Heun's Method)

$$k_1 = f(u^n)$$

$$k_2 = f(u^n + \Delta t \cdot k_1)$$

$$u^{n+1} = u^n + \frac{\Delta t}{2}(k_1 + k_2)$$

Error: $O(\Delta t^2)$

RK2 (Midpoint Method)

$$k_1 = f(u^n)$$

$$k_2 = f\left(u^n + \frac{\Delta t}{2}k_1\right)$$

$$u^{n+1} = u^n + \Delta t \cdot k_2$$

RK4 (Classical)

$$k_1 = f(u^n)$$

$$k_2 = f\left(u^n + \frac{\Delta t}{2}k_1\right)$$

$$k_3 = f\left(u^n + \frac{\Delta t}{2}k_2\right)$$

$$k_4 = f(u^n + \Delta t \cdot k_3)$$

$$u^{n+1} = u^n + \frac{\Delta t}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

Error: $O(\Delta t^4)$

Amplification:

$$G(z) = 1 + z + \frac{z^2}{2} + \frac{z^3}{6} + \frac{z^4}{24}$$

Leap-Frog Method

$$u^{n+1} = u^{n-1} + 2\Delta t \cdot f(u^n)$$

- **Error:** $O(\Delta t^2)$
- **Starting problem:** Needs u^0 and u^1

BDF (Backward Differentiation)

$$\frac{\partial u}{\partial t} \approx \frac{3u^{n+1} - 4u^n + u^{n-1}}{2\Delta t}$$

KEY FORMULAS & TRICKS

Complex Exponentials

$$|e^{i\theta}| = 1 \quad (\text{always!})$$

$$|e^{a+ib}| = e^a$$

Small Angle Approximations

$$\sin\left(\frac{k\Delta x}{2}\right) \approx \frac{k\Delta x}{2} \quad \text{as } \Delta x \rightarrow 0$$

$$\sin^2\left(\frac{k\Delta x}{2}\right) \approx \frac{(k\Delta x)^2}{4}$$

Common Stability Constraints

$$\text{Heat (Explicit): } \Delta t \leq \frac{\Delta x^2}{2\nu}$$

$$\text{Euler Explicit: } -2 \leq \lambda \Delta t \leq 0$$

Method of Undetermined Coefficients

To derive FD formula for $\frac{\partial^k u}{\partial x^k}$:

1. Write: $\left. \frac{\partial^k u}{\partial x^k} \right|_j = \sum_i a_i u_{j+i}$
2. Taylor expand each u_{j+i}
3. Match coefficients: $u, u', u'', \dots$
4. Solve for $\{a_i\}$

Method	Order	Stability	Cost
Euler Explicit	$O(\Delta t)$	Conditional	Low
Euler Implicit	$O(\Delta t)$	Unconditional	Medium
Trapezoidal	$O(\Delta t^2)$	Unconditional	Medium
RK2	$O(\Delta t^2)$	Conditional	Medium
RK4	$O(\Delta t^4)$	Conditional	High
Leap-Frog	$O(\Delta t^2)$	Conditional	Low

Table 1: Time Integration Methods Comparison

PROBLEM-SOLVING CHECKLIST

Stability Analysis

1. Substitute Fourier mode: $u = e^{i(kx+\omega t)}$
2. Get amplification factor G
3. Check $|G| \leq 1$
4. Solve for time step constraint

Deriving FD Formulas

1. Write Taylor expansions
2. Add/subtract to eliminate unwanted derivatives
3. Solve for desired derivative
4. Identify leading error term

PDE Classification

1. Write: $Au_{xx} + Bu_{xy} + Cu_{yy} = \dots$
2. Calculate $\Delta = B^2 - 4AC$
3. Check sign of Δ

COMMON PITFALLS

- **Stability $\neq$ Accuracy**
- **CFL Conditions:**
 - Diffusion: $\Delta t \propto \Delta x^2$ (more restrictive!)
 - Advection: $\Delta t \propto \Delta x$
- Always check $\text{Re}()$ for growth
- Don't forget boundary conditions
- Multi-step needs multiple initial values

CALCULUS REVISION

Derivatives (42)

Basic:

$$\frac{d}{dx}(x^n) = nx^{n-1}, \quad \frac{d}{dx}(e^x) = e^x$$
$$\frac{d}{dx}(\sin x) = \cos x, \quad \frac{d}{dx}(\cos x) = -\sin x$$

Chain: $\frac{d}{dx}[f(g(x))] = f'(g) \cdot g'$

Product: $\frac{d}{dx}[fg] = f'g + fg'$

Integrals (43)

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C, \quad \int e^x dx = e^x + C$$

FTC: $\int_a^b f'(x) dx = [f(x)]_a^b = f(b) - f(a)$

Partial Derivatives (44)

$\frac{\partial u}{\partial x}$ = derivative w.r.t. x (others constant)

Example: $u = x^2y + 3t$

$$\frac{\partial u}{\partial x} = 2xy, \quad \frac{\partial u}{\partial y} = x^2, \quad \frac{\partial u}{\partial t} = 3$$

Higher: $\frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial x} \right)$

Key PDE Techniques (45)

Leibniz: $\frac{d}{dt} \int_a^b f(x, t) dx = \int_a^b \frac{\partial f}{\partial t} dx$

Total derivative in integral:

$$\int_a^b \frac{\partial F}{\partial x} dx = [F]_a^b$$

Product rule for partials:

$$\frac{\partial}{\partial x}(uv) = \frac{\partial u}{\partial x}v + u \frac{\partial v}{\partial x}$$

Key identity: $u \frac{\partial u}{\partial x} = \frac{1}{2} \frac{\partial}{\partial x}(u^2)$

GOOD LUCK!